

WASP-2b

R-2185

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-2_101009 · created 2026-08-13T23:18:03+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2455478.6981 +/- 0.0013 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.124 +/- 0.011
Transit depth (Rp/Rs)^2	1.53 +/- 0.28 [%]
Orbital Inclination (inc)	84.7 +/- 0.41
Airmass coefficient 1 (a1)	1.2153 +/- 0.007
Airmass coefficient 2 (a2)	-0.1266 +/- 0.0038
Scatter in the residuals of the lightcurve fit is	0.58 %
Best Comparison Star	None
Optimal Aperture	2.67
Optimal Annulus	6.42
Transit Duration (day)	0.0714 +/- 0.0048

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.124 ± 0.011 vs 0.1326 (deviation 0.78σ)
Duration fit vs published	0.0714 d vs 0.0737 d (deviation 0.49σ)
Mid-time O–C	-3.9 min
Depth SNR	11.3
Residual scatter	0.58 %

Light curve

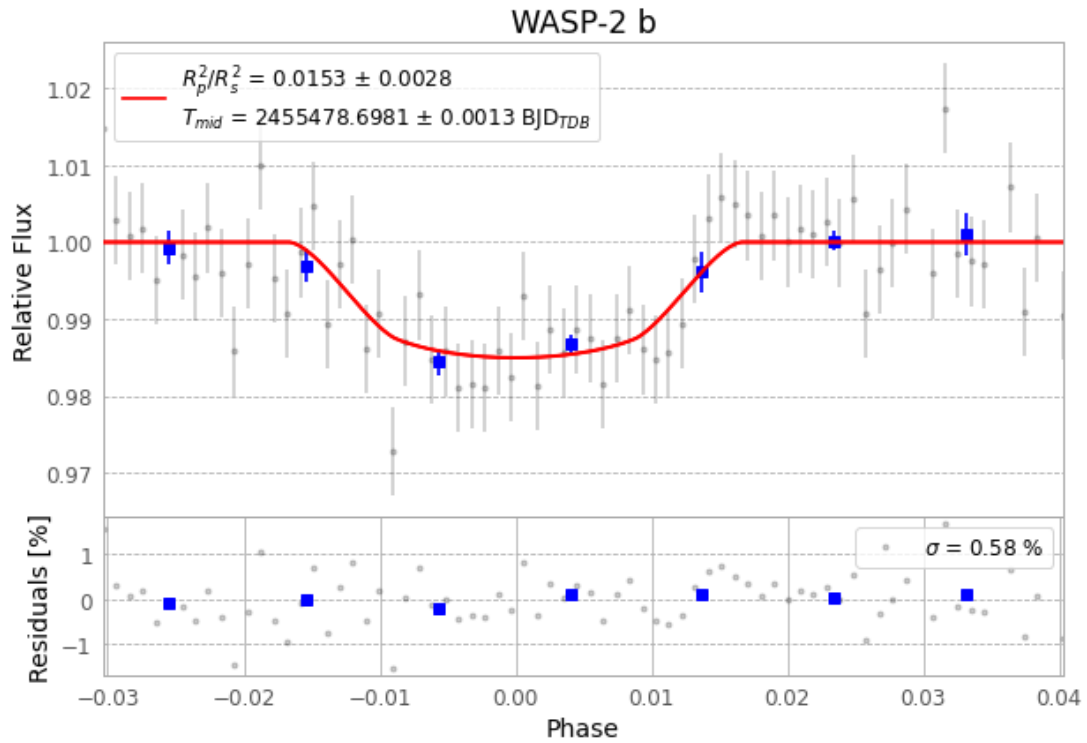


Figure 1 — FinalLightCurve_WASP-2 b_2010-10-08.png (SHA-256 260a86237dce4010...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [193, 218], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 2b298f64aa2282fb...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

74 FITS frames (49 MB), WASP-2101009030614.FITS → WASP-2101009064515.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-2-101009.log (f7cd0fd99689...), FinalLightCurve_WASP-2 b_2010-10-08.png (260a86237dce...), FinalLightCurve_WASP-2 b_2010-10-08.csv (5efe5589f7f3...)

Compute resources

Wall time 123.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-13 23:18Z.